

Unit 11: Worked Examples

Find the sum of each expression.

1. $-\frac{2}{5} \cdot (-7\frac{1}{3})$

$$-\frac{2}{5} \cdot (-\frac{22}{3}) = \frac{44}{15}$$

$$- \cdot - = +$$

2. $-243 \div 40.5$

$$40.5 \overline{) -243} \begin{array}{r} -6 \\ \hline \end{array}$$

$$- \div + = -$$

Solve the following word problems.

3. Julianna is training for a 10k and ran 8.2 miles on Monday, 2.9 miles on Tuesday, and 6.7 miles on Thursday. Her friend Max is also training and he ran $\frac{3}{5}$ of Julianna's total mileage. How many miles did Max run?

$$\begin{array}{r} \text{Julianna} = 8.2 \\ + 2.9 \\ + 6.7 \\ \hline 17.8 \end{array}$$

$$\text{max} = \frac{3}{5} (17.8)$$

$$\frac{3}{5} \cdot \frac{17.8}{1} = \frac{53.4}{5} = 10.68 \text{ miles}$$

4. Cecilia has a \$55 gift card to her local game store. She has a VIP membership, which gives her a 35% discount off any item. Cecilia is wanting to buy a new game pack that costs \$52.40. How much will be left on her gift card after her purchase?

$$52.40(0.65) = 34.06$$

$$\begin{array}{r} 55.00 \\ - 34.06 \\ \hline \$20.94 \end{array}$$

5. Andre is making a diorama of a farm for his agriculture class. He has 29.15 inches of wire to use for fencing in the diorama and 12 toothpicks to use as fence posts. Andre wants to make sure that the toothpicks are spaced evenly across the wire. How much wire needs to go between each fence post?

$$\begin{array}{cccccccccccc} | & | & | & | & | & | & | & | & | & | & | & | \\ \checkmark & \checkmark & \checkmark & \checkmark & \checkmark & \checkmark & \checkmark & \checkmark & \checkmark & \checkmark & \checkmark & \checkmark \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & \end{array}$$

$$29.15 \div 11$$

$$\begin{array}{r} 265 \\ 11 \overline{) 2915} \\ \underline{22} \\ 71 \\ \underline{66} \\ 55 \end{array}$$

$$265$$

$$2.65 \text{ inches}$$